LAB Exam 2nd BATCH

Reg. No. | | | | | | | | | | | [13CEL69/12CEL69]

SIXTH SEMESTER B.Tech./B.E EXAMINATION, JUNE – 2016

DESIGN AND DRAWING OF STRUCTURES

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Assume suitable non trivial data wherever necessary.
2. Use of IS456:2000 is allowed. Use of SP-16 is allowed.
3. Answer any **ONE** full Question from each part A and part B

PART - A

1. a. Design a T-Shaped container retaining wall to retain earth embankment 3m high above ground level. The unit weight of earth is 18KN/m^3 & the angle of repose is 30^0. The embankment is horizontal at top. The SBC of soil may be taken as 100KN/m^2 & the co-efficient of friction between soil & concrete as 0.5 use M20 mix & Fe-415 HYSD bars. Draw following sections showing the details of reinforcement. **50**

   i. Cross section of wall.

   ii. Longitudinal section of stem.

   iii. Longitudinal section of base slab at top & bottom.

2. a. The roof of a 8m wide hall is supported on a portal frame spaced at 4m intervals. The height of the portal frame is 4m the continuous slab is 120mm thick. Live load on roof=1.5KN/m^2. **50**

   Bearing capacity of soil=150KN/m^2.

   Columns are connected with a plinth beam & the base of the column may be assumed as fixed. Design the column & beam members & a suitable foundation footing for the columns of the portal frame. Adopt M20 grade concrete & Fe415 HYSD bars.

   Draw following sections showing the details of reinforcement.

   i. Cross section of continuous slab.

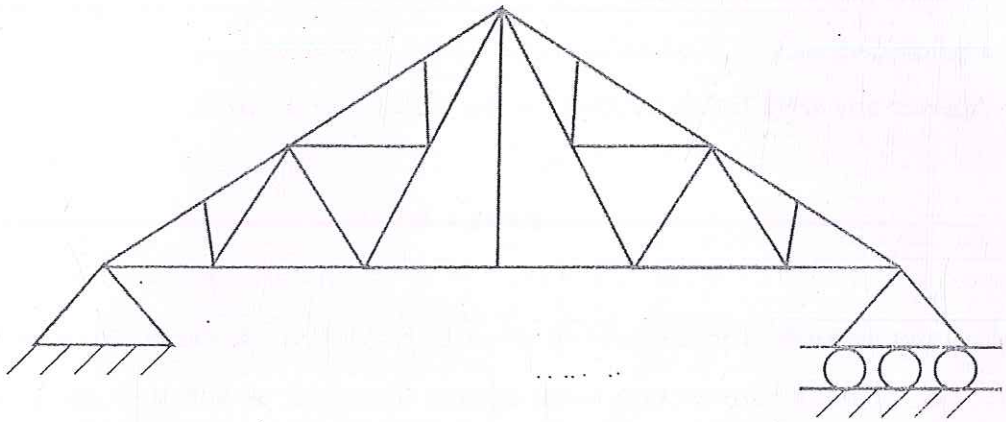
   ii. Longitudinal section of column.

   iii. Cross section of footing.

PART - B

3. a. Draw a fink roof truss shown in figure below for the data given: 20

Affix the notations



Span=24m

Rise=6m

Principle rafter: two unequal angles ISA 100mmx75mmx8 mm

Principle tie: Two equal angles ISA 75mmx75mmx8 mm

Struts & all other member: ISA 55mmx55mmx8 mm Gusset plates: 10mm thick.

Show the connection details at ridge & support in separate drawing. All connections are bolted.

4. a. Draw a plate girder for the following data **20**

span=24m

Web plate: 2200mmx10mm

Flange plates: 560mmx50mm

End bearing stiffener: 190mmx20mm

Intermediate stiffener: 220mmx20mm

Spacing of intermediate stiffener: 2666.66mm

All connections are shop convex fillet weld of size 5mm.

Draw C/S, L/S & top view.